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Competency-based Curriculum Mapping as a Tool for Continuous Improvement for Master of Public Administration (MPA) Programs

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ABSTRACT

Curriculum mapping (CM) is an assessment tool used to articulate and revise an academic curriculum. Network of Schools of Public Policy, Affairs and Administration (NASPAA) accredited graduate programs in public policy, public affairs, and public administration are required to demonstrate a competency-based curriculum development and assessment methods. This article provides the rationale and approach used to generate a competency-based Master of Public Administration (MPA) curriculum and describes a systematic approach to incorporating program learning objectives, stakeholder engagement, competency development or revisions by faculty, into competency-based CM. The competency-based curriculum is relatively new for MPA programs. While there are many potential benefits and challenges, this article focuses on the utility of flexible mission-oriented curriculum design and its ability to link competencies to assessment strategies for MPA programs.

KEYWORDS

Competencies; competency-based curriculum mapping; MPA; professional education; program assessment

Introduction

With the increased attention to assessment among higher education institutions, more institutional resources are now being allocated to design innovative and effective methods of evidence-based assessment. Curriculum mapping (CM) is considered an effective tool for improving teaching and learning (Jacobs, 2004) to ascertain whether a program's courses and educational content are addressing specified competencies (American Educational Research Association, American Psychological Association, & National Council on Measurement in Education, 1999; Awortwi, 2010; Bath, Smith, Stein, & Swann, 2004; Ervin, Carter, & Robinson, 2013; Glass, 2007; Liu, Wrobbel, & Blankson, 2010). It considers when and what is taught, as well as assessment strategies used to examine the achievement of specific learning outcomes for academic degree programs (Harden, 2001; Spencer, Riddle, & Knewstubb, 2012). Curriculum maps draw explicit connections between learning objectives, experiences, and outcome assessment (Drake & Burns, 2004). Several benefits of CM have been identified through helping faculty identify alignment, gaps, overlaps, inconsistencies, and strengths within a program (Allen, 2004; Jacobs, 2004; Kapucu, 2011).

CM can be defined as examining the key elements of curriculum with outcome objectives to identify successes, potential gaps, and redundancies to improve the

effectiveness of teaching and learning in an academic program. CM can be presented in forms such as text, tables or flow charts comparable to geographic mapping in its ability to visually link resources with outcomes (Ervin et al., 2013; Harden, 2001; Liu et al., 2010). In addition, CM has been used to demonstrate adherence to accrediting agency standards (Iglar, Whitehead & Takahashi, 2013; Kinchin, 2011; Spencer et al., 2012; Sumsion & Goodfellow, 2004). Competency-based curriculum issues have been addressed for programs closely related to public administration such as nonprofit management (Wang & Ashcraft, 2012) and emergency management (Kapucu, 2011).

The effectiveness of academic degree programs is regularly requested by the accreditation agencies. The Commission on Colleges of the Southern Association of Colleges and Schools (SACS), 2005 states:

[curriculum] coherence is a critical component of a program and should demonstrate an appropriate sequencing of courses, not a mere bundling of credits, so that student learning is progressively more advanced in terms of assignments and scholarship required and demonstrates progressive advancement in a field of study that allows students to integrate knowledge and grow in critical skills. (p. 12)

The Association of American Colleges and Universities (AACU) encourages faculty engagement in assessment of student learning outcomes at the program level.

The Network of Schools of Public Policy, Affairs and Administration (NASPAA) has been accrediting master's level programs on public policy, public administration, and public affairs for more than 30 years and has been encouraging the competency-based curriculum, but not officially mandating it until 2009 for Master of Public Administration (MPA) programs¹. Public administration education accreditation standards have been recently shifted from objective-based (input-output) standards to a competency-based (outcome and performance-oriented) approach (Henry, Goodsell, Lynn, Stivers, & Wamsley, 2009; Mau, 2015; Raffel, Maser, & Calarusse, 2012; Rubai & Calarusse, 2012). One distinguishing characteristic of a competency-based curriculum is the focus on mastery of a specific set of capabilities for graduates from MPA programs. Table 1 includes the NASPAA's universal required competencies and the University of Central Florida (UCF) MPA program learning objectives. In response to these requirement changes, accredited MPA programs reorganize or revise curriculum and assessment strategies and plans.

The article is a case study of the MPA CM exercise undertaken by the UCF MPA program and that this case ostensibly presents some lessons for other MPA programs going through a similar process in the United States and around the globe. This article provides background information on competency-based curriculum, CM, competency-based MPA CM, and outcome assessment strategies. In addition, examples are provided for the competency-development process along with aligning teaching and assessment to program-level competencies with respect to NASPAA accreditation standards. Discussion and recommendations on CM and developing rubrics for assessment strategies also are included. Since it is relatively new, the CM process for the MPA programs needs more cases. As more research is conducted among NASPAA member institutions, it is anticipated the process of CM to competencies (and learning objectives) can be streamlined and made less resource intensive. The first part of the article provides a boarder perspective on CM, the second part provides an example and lessons learned from an MPA program.

Competency-based curriculum and CM

NASPAA requires implementation of five core competencies within its accreditation standards for MPA programs for student learning and career preparedness for public service (Table 1). These competencies are critical in articulating specific skill sets needed by public administrators

Table 1. NASPAA universal required competencies and UCF MPA program learning.

Objectives
To lead and manage in public governance.
Develop expertise on local government.
Develop an ability to lead and manage within multiple networks of stakeholders.
Reinforce democratic principles and practices.
Produce consensus or consent among diverse stakeholders.
Model the skills and power of leadership (including transformational, transactional, and servant) and utilize principles of public management.
To participate in and contribute to the policy process.
Develop capacity to analyze political theory, policy process, and implementation.
Analyze policy issues/problems using different methodologies.
Identify and evaluate interests from multiple stakeholders in policy formation, and communicate evidence-based policy information as a foundation for collaborative negotiations with diverse stakeholders.
Assess the political, legal, economic, institutional, and social environment and operation of public administration in the policy process (including concepts, theories, approaches, models, techniques, formulation, implementation, evaluation, and institutions).
Design policy implementation methodology and actively engage with public sector employees and citizens to develop goals and strategies.
To analyze, synthesize, think critically, solve problems and make decisions
Develop expertise in techniques used in managing government operations.
Appraise the concepts of social science research methods, statistical analysis and techniques; apply the skills to analyze public policies and government operations; and formulate a policy or managerial decision.
Create networked/collaboration solution to complex modern problems.
Identify, evaluate, and model best practices.
Develop professional capacity in basic skills (including writing, speaking, analytical techniques, and critical thinking).
To articulate and apply a public service perspective.
Develop a respect and demonstrate methods for authentic interaction with citizens, other governmental entities, nonprofit and business communities.
Value and demonstrate commitment to professionalism and integrity in serving the public.
Communicate public interest based on ethical reasoning and democratic participation.
Critique instrumental reasoning in order to promote social and economic equity and justice.
Incorporate and value principles of democracy, public transparency, and consensus building in the workplace when making decisions that directly impact the community.
To communicate and interact productively with a diverse and changing workforce and citizenry.
Incorporate various communication tools and strategies (including written reports, video technologies, brochures, emails, PowerPoint, social media, and marketing plans) in the management of public organizations.
Incorporate major concepts, skills, process, and policies (including planning, classification, compensation, recruitment, selection, training, performance appraisal, labor relations, equal employment opportunity, and affirmative action) in public service human resource management.
Develop cultural competency and appreciation of diversity for collaborating with diverse stakeholders to produce consensus or consent.
Negotiate interest-based resolutions with stakeholders experiencing conflict.
Critique existing partnerships and promote engagement and interaction with citizens, nonprofit organizations, and collaborative efforts with other entities.

and creating quality standards for institutions and educators. Competency-based or standard-based curriculum and education has been promoted since the early 70s with

¹For additional information please visit NASPAA accrediting webpage at <http://www.naspaa.org/accreditation/NS/about.asp>.

curriculums focusing on competencies or capabilities to be attained by the students rather than on prescribed content (Awortwi, 2010; Spencer et al., 2012; Williams & Bolland, 2011). The competency-based curriculum is particularly common in clinical professions, such as psychology, nursing, medicine, and social work (Klein, 2006; Litzelman & Cottingham, 2007).

Even though the competency-based curriculum provides flexibility to faculty in designing course content, it is not free of criticism. Some critics believe that this method is more appropriate for vocational education rather than academic programs not who traditionally focus on abstract, critical, and analytical thinking skills (Ashworth & Saxton, 1990).

CM can ensure competencies are covered in the courses and are an essential element of assessment processes for academic programs (Figure 1). A CM activity places teaching and learning activities in the center. Assessing program level competencies can include direct and indirect measurements. The model also highlights how the learning outcomes play an essential role in determining learning objectives and activities in an academic program. Direct measures involve, for example, projects completed by students and graded by a systematic rubric (Sandoval & Wigle, 2006). Indirect measures involve, for example, exit or graduation surveys along with personal reflections (Martell & Calderon, 2005). Simply relating indirect measures of educational experience and learning are not sufficient for outcome assessment (Wesp & Miele, 2008). Some combinations of direct and indirect assessment strategies are best for curriculum and program effectiveness measures. At the UCF MPA program, competency-based portfolio, capstone projects, and graduating student surveys are being utilized as assessment strategies.

CM includes two key functions: 1) it makes the curriculum more transparent to all the stakeholders involved

and 2) it demonstrates the links between the different elements or learning and teaching for an academic program (Harden, 2001, p. 124). The curriculum map also allows for identifying gaps in addressing effective teaching and learning in addressing competencies identified for a program (Sumsion & Goodfellow, 2004). Williams and Bolland (2011) highlighted that CM is not merely a reiteration of the curriculum, but rather a reconstruction of the essential components through prescriptive and descriptive approaches. Embodying concepts of intentionality and transparency, the results of the method outline how institutional goals are bridged with those of the educational content (Cuevas, Matveev, & Miller, 2010; Harden, 2001). Norfolk State University (NSU) utilizes CM for its complex dynamic system and interdependent components, for example. Intentionality is outlined in the courses' interaction effects on student development and learning (Cuevas et al., 2010). Transparency in the program curriculum maps, exhibited in the NSU case, showcase the need for continual enhancement of programs and creation of a collegial, supportive environment. Not to mention, the transparency allows faculty and staff to discern the influence of cognitive processes they inspired (Cuevas et al., 2010; Kinchin, 2011). In-depth elements of a curriculum map is presented in Figure 2. The figure highlights the relationship between the structure of the discipline and curriculum (content of coverage may be sequential or not); structure of the curriculum (the importance of pedagogical content knowledge [PCK]), and the structure of student learning and outcome measure). The figure also demonstrates the link between the conceptual and experiential components of the curriculum.

Similar uses of CM are identified within international institutions of higher education as well (Bath et al., 2004; Ervin et al., 2013; Iglar et al., 2013). Australian universities utilize CM in developing programs and measuring

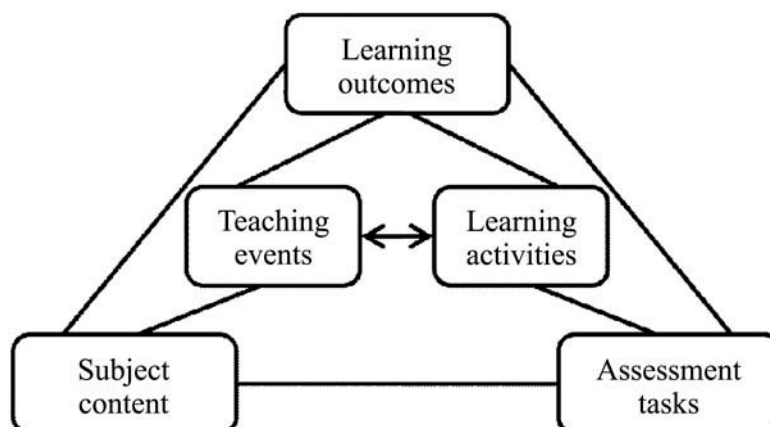


Figure 1. Key elements of curriculum mapping method (Bester & Scholtz, 2012, p. 293).

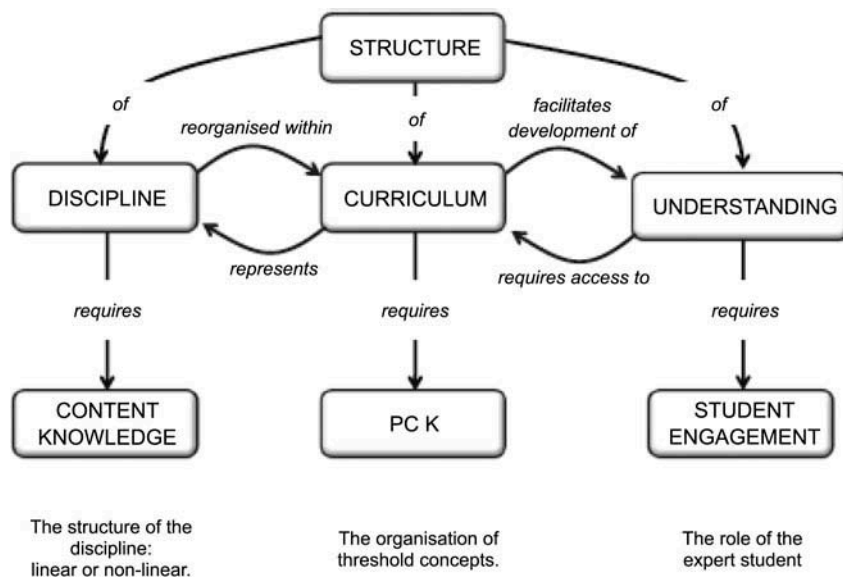


Figure 2. In-depth elements for curriculum map (Kinchin, 2011).
Note. PCK: pedagogical content knowledge.

the outcomes (Bath et al., 2004; Liu et al., 2010). South Africa sees the tool as a way to critically examine the responsiveness and relevance of higher education in terms of the “intended (planned), the taught (created) and the learned (experienced)” outcomes (Bester & Scholtz, 2012, p. 283). Specifying the incorporation of stakeholder analysis into a constructivist framework, universities promote cognitive complexity for students to thoroughly understand and apply their knowledge in activities “such as interpreting, summarizing or explaining” (Bester & Scholtz, 2012, p. 288).

Regardless of location, the mapping process has a tendency to bring about more questions than answers necessitating the support of faculty and staff to impact the method’s effectiveness (Holycross, 2006). Due to the time-consuming and longitudinal nature of CM, committed involvement is a challenge (Iglar et al., 2013; Kinchin, 2011; Liu et al., 2010; Uchiyama & Radin, 2009). Dedicated support provides a footing for implementing changes to advance educational content (Hall & Hord, 2006; Harden, 2001; Iglar et al., 2013; Williams & Bolland, 2011) and focus on quality assurance issues and assessment strategies (Bath et al., 2004). A potential reason is the variance of “institutional processes, cultures, and context-specific issues” (Sumsion & Goodfellow, 2004, p. 331).

If intentionality and transparency are promoted, then competency-based knowledge can be productively managed (Cuevas et al., 2010; Harden, 2001) and, more importantly, tracked for accountability purposes (Holycross, 2006). In a joint statement by the Council for Higher Education Accreditation (CHEA) and the Association of American College and Universities

(AAC&U) provide the overarching goal of higher education: “we in higher education must constantly monitor the quality of student learning and development, and use the results both to improve achievement and to demonstrate the value of our work to the public” (Liu et al., 2010, p. 238). Since competency-based curricula are becoming increasingly popular among public service professions, emergent outcomes consist of increased accountability and adherence to the standards of relevant accrediting bodies (Williams & Bolland, 2011).

Portfolios, in addition to other activities in an academic program, could be utilized as a tool to align the competency expectations of the accrediting bodies with the program structure. In particular, the portfolio assessment method allows for the faculty to maintain the program’s relevancy and validate the “goals, training inputs, roles, tasks, competencies, skills, objectives, and specific performance statements” (Bawane & Spector, 2009, p. 384).

Curriculum development and mapping are considered a bottom-up approach with faculty and staff engagement being critical (Williams & Bolland, 2011). However, it is difficult to bring faculty on board due to the focus on research for tenure and promotion rather than teaching and service (Baker et al., 2007; Kuh & Ikenberry, 2009). Sometimes, the importance of program assessment strategies discussed by university administrators but assessment focuses on program management rather than student competencies and programmatic outcomes (Tilbury & Wortman, 2008). These have been discussed on course-based improvement, such as curriculum and syllabus development (Donald, 1997;

Earley, 2007; Suddreth & Galloway, 2006), and collaborative teaching and learning (Barkley, Cross, & Major, 2005; Kapucu, 2012; Koliba, 2004). There is limited information available on the process of developing competency-based curriculum, and linking the course learning objectives to program effectiveness and outcome assessment strategies, especially professional degree programs such as MPA.

Benefits and challenges of CM

The benefits of CM include quality assurance, benchmarking, accountability, alignment of institutional goals to curriculum outcomes, transparency to accreditation standards, faculty and staff engagement, and promotion of cognitive process (Bath et al., 2004; Cuevas et al., 2010; Holycross, 2006; Iglar et al., 2013; Kinchin, 2011; Liu et al., 2010; Robley, Whittle, & Murdoch-Eaton, 2005b; Williams & Bolland, 2011). Additional benefits include opportunity to establish patterns, reflect and dialogue about strategic development, and faculty engagement (Damrosch, 1995; Sumsion & Goodfellow, 2004). The map can be beneficial for the faculty who teaches the program, students, advisory board, potential employers, and newly hired faculty orientations. Students and faculty will be able to see the role of individual courses in the program as well as the big picture of the entire program as a complete package contribution professional development of public service leaders.

Challenges of the mapping process include the time necessary for completion, implementation and continued assessment along with definitions of competencies and outcomes (Bath et al., 2004; Cuevas et al., 2010; Holycross, 2006; Iglar et al., 2013; Liu et al., 2010). Additionally, Ervin et al. (2013) highlight discrepancies in philosophical interpretations, difficulties in determining proficiency levels, background of rater, and relationships between competencies. Therefore, clarification is an essential component for reliable outcomes of competency-based education, which is a key element to ascertaining critical skill sets of public service professions and professional degree programs (Ervin et al., 2013).

Implementation

Harden (2001) detailed a generic five-step process beginning with *assessing the needs* and understanding the reason behind the curriculum map itself. Then, the

next step is to *scope the task* in terms of complexity and the levels to be included. Once levels are determined, then it is time to *establish the links* between the levels before determining what information is needed to assess the programs and who will be responsible for collecting and examining the data, which is the fourth step of *populating the windows*. The last step is to *decide the format for the map* to represent the best snapshot of the results. The presentation is imperative to convey the information to the identified stakeholders (Holycross, American Educational Research Association, American Psychological Association, & National Council on Measurement in Education, 1999; 2006; Kinchin, 2011). Table 2 provides visual examples of CM, course sequence, and competency-based learning outcomes.²

Narrowing the focus, Liu et al. (2010) gave three mechanisms to align curriculums: In individual courses: Each course uses assessment tools, such as exams or research papers to assess learning; In higher-level courses: When a lower level course is a prerequisite for a higher-level course, the higher-level course can be an assessment tool for learning that takes place in the lower-level course; and In the capstone experience course: The capstone experience course, if one exists, which students take at the end of their program, serves as the cumulative assessment tool (p. 242). The three perspectives are being implemented at UCF MPA programs as demonstrated in Table 2. The table and the core elements are explained in the second part of the paper in details.

Additional questions to consider revolve around course sequences along with establishing appropriate balance and connections (Liu et al., 2010). Uchiyama and Radin (2009) also stated a distinction of CM being in real time, which “in this context means when the curriculum is delivered, rather than as projected in a course syllabus prior to the course or after the course is completed” (p. 272).

Differed tolls and methods can be used for CM. Through the use of a facilitator and a well-equipped space (e.g. whiteboards, post-it notes, paper, markers, etc.), faculty and staff can evaluate a program’s curriculum and course sequence to ease the process. Understanding the importance of a skilled facilitator, Hale and Dunlap (2010) highlighted the CM process in relation to the administrative leaders who will guide the dialogue. The hope is to enhance leadership skills to sustain creative and collaborative environments. Udelhofen (2005) integrated the process of mapping with the importance of generating a creative

²The curriculum map includes only core courses and mastery level for at least one universal competency. This can be revised and add for other learning objectives. Adding elective courses to the curriculum map can also be considered in the future with matching specific learning outcomes.

Table 2. NASPAA universal competencies and MPA program learning objectives.

MPA core courses	Competencies and program learning objectives				
	1	2	3	4	5
PAD 6053 Public Administrators in the Governance Process	I	I	R	I	R
PAD 6035 Public Administration in the Policy Process	M/A	M	M	R	R
PAD 6700 Research Methods in Public Administration		R	M/A		R
PAD 6037 Public Organization Management	M/A	R	R	I	R
PAD 6701 Analytic Techniques for Public Administration		R	M/A		R
PAD 6227 Public Budgeting	M		M/A		
PAD 6207 Public Financial Management		R	M/A		R
PAD 6417 Human Resource Management			M	R	M/A
PAD 6335 Strategic Planning and Management	M	R	M	R	M/A
PAD 6062 Advanced Concepts and Applications in Public Administration	M/A	M	M/A	M	M/A
MPA Core Courses (lead and manage in public governance)	<i>Competencies and program learning objectives</i>				
	1a	1b	1c	1d	1e
PAD 6053 Public Administrators in the Governance Process		I	I	I	I
PAD 6035 Public Administration in the Policy Process	I		R	M	
PAD 6700 Research Methods in Public Administration					
PAD 6037 Public Organization Management	R	M	I	R	M
PAD 6701 Analytic Techniques for Public Administration					
PAD 6227 Public Budgeting	M	I		I	
PAD 6207 Public Financial Management					
PAD 6417 Human Resource Management					
PAD 6335 Strategic Planning and Management	M	R	R	M	R
PAD 6062 Advanced Concepts and Applications in Public Administration		M	R	M	M
MPA Core Courses (participate in and contribute to the policy process)	<i>Competencies and program learning objectives</i>				
	2a	2b	2c	2d	2e
PAD 6053 Public Administrators in the Governance Process	I		I	I	I
PAD 6035 Public Administration in the Policy Process	R	I	R	M	I
PAD 6700 Research Methods in Public Administration	R	I	I	I	I
PAD 6037 Public Organization Management			R	R	
PAD 6701 Analytic Techniques for Public Administration	R	R	R	R	R
PAD 6227 Public Budgeting					
PAD 6207 Public Financial Management		R	R	R	
PAD 6417 Human Resource Management					
PAD 6335 Strategic Planning and Management		R	R	R	R
PAD 6062 Advanced Concepts and Applications in Public Administration	R	R	M	R	R
MPA Core courses analyze, synthesize, think critically, solve problems and make decisions)	<i>Competencies and program learning objectives</i>				
	3a	3b	3c	3d	3e
PAD 6053 Public Administrators in the Governance Process	I		I		R
PAD 6035 Public Administration in the Policy Process	M		R		R
PAD 6700 Research Methods in Public Administration	I	M		R	M
PAD 6037 Public Organization Management	R	I	I	R	R
PAD 6701 Analytic Techniques for Public Administration		M		R	R
PAD 6227 Public Budgeting	M			R	R
PAD 6207 Public Financial Management	M	R			R
PAD 6417 Human Resource Management	M			I	R
PAD 6335 Strategic Planning and Management	R	R	R	M	R
PAD 6062 Advanced Concepts and Applications in Public Administration	R	R	M	R	M
MPA Core courses (articulate and apply the public service perspective)	<i>Competencies and program learning objectives</i>				
	4a	4b	4c	4d	4e
PAD 6053 Public Administrators in the Governance Process	I		I	I	I
PAD 6035 Public Administration in the Policy Process	R		R	R	R
PAD 6700 Research Methods in Public Administration					
PAD 6037 Public Organization Management	I	I			I
PAD 6701 Analytic Techniques for Public Administration					
PAD 6227 Public Budgeting					
PAD 6207 Public Financial Management					
PAD 6417 Human Resource Management		R			
PAD 6335 Strategic Planning and Management	R		R	R	R
PAD 6062 Advanced Concepts and Applications in Public Administration	M	M	R	R	R
MPA Core courses (communicate and interact productively with a diverse and changing workforce and citizenry)	<i>Competencies and program learning objectives</i>				
	5a	5b	5c	5d	5e
PAD 6053 Public Administrators in the Governance Process			R	I	I
PAD 6035 Public Administration in the Policy Process	R		R	R	R
PAD 6700 Research Methods in Public Administration	R				
PAD 6037 Public Organization Management	I	I	R	R	
PAD 6701 Analytic Techniques for Public Administration	R				
PAD 6227 Public Budgeting					
PAD 6207 Public Financial Management	R				
PAD 6417 Human Resource Management		M	R		
PAD 6335 Strategic Planning and Management	M		M	M	M
PAD 6062 Advanced Concepts and Applications in Public Administration	M	R	M	R	M

environment which can be a useful to organize a set of workshops for faculty and staff.

Similar to what was discussed above, competency-based learning objectives stipulate the knowledge and skills students should possess, according to NASPAA, upon completion of MPA program. MPA program competencies are measured at various developmental levels: Introduced, Reinforced, and Mastered. These three categories adapted for the MPA program identified as declared, taught, and learned by English (1978) earlier in the literature.

CM at the university of central Florida MPA program

The School of Public Administration at the University of Central Florida (UCF) uses CM as a way of showing when and where, within a program, a student gains exposure to the primary competencies, knowledge, and skills of the discipline and existing opportunities to demonstrate their level of attainment. The major projects, where mastery of a competency is measured through a grading rubric, become the “direct measures” for the evaluation for the Institutional Effectiveness (IE) program³. Faculty members teach in the MPA program come to an agreement regarding the assignment and grading rubric used to capture mastery of the competency. Then these “direct measures” are used and graded, which allow faculty teaching in the MPA program to measure, in a consistent way, how well their students master the competencies. An example of a CM for the MPA program is provided below using the terms of: introduced (I), reinforced (R), and mastered (M).

Introduced (I)

At this level, courses introduce the concepts relevant to the program competences and activities focus on basic knowledge and skills, which support the student’s learning outcome. Only one, or a few, aspect(s) of the program competencies may be addressed in the given course; however, students are exposed to at least one element, which serves as a building block toward full achievement later in the curriculum. In a required research methods course, for example, students are introduced to the basics of research methods, and complete an in class assignment and participate in discussions of the new concepts based on their newly acquired foundational knowledge.

Reinforced (R)

At this level, courses reinforce or contribute to opportunities for achieving a competency and afford students opportunities to practice. The course strengthens, supports, and *reinforces* the development of knowledge and skills necessary for optimal achievement in latter courses as the foundational information of the competency was introduced in previous course work. Typically, several aspects are addressed and activities continue with increased complexity. At this level, there may not be a holistic integration of all the knowledge, skills, and attitudes necessary for full achievement of the program competency. In a required course, for example, students practice their knowledge of research methods by completing a proposal development assignment.

Mastered (M)

At this level, courses promote mastery through emphasizing opportunities for students to integrate all the knowledge, skills, and attitudes necessary. Instructional and learning activities focus on applying skills tied to the competency in multiple contexts and complexity. The competency was introduced and reinforced through previous courses in the curriculum. Ideally, students had sufficient practice and can demonstrate mastery. The students present their own knowledge as part of a capstone project or a completion of a major course research paper assignment utilizing research methods and statistical analysis, for example.

MPA program CM example

The curriculum map, showcasing competencies and learning objectives matched with introduced (I), reinforced (R), and/or mastered (M), for the MPA program is included in a table. Robley, Whittle, and Murdoch-Eaton (2005a, 2005b) addressed the three categories we used in mapping the curriculum for our MPA program at UCF, and add *assessed* as a fourth category. Under this category, evidence is collected and evaluated for program level assessment as part of the CM. If preferred an (A) can be added in the curriculum map indicating when the data is collected. Data for assessment purposes can be collected at every levels identified by *R*, *M*, or *I* but this can be time consuming, challenging, and complicated. Evidence can also be collected

³University of Central Florida (UCF) uses Institutional Effectiveness (IE) measure for its academic programs for Organizational Excellence and Assessment Support (OEAS) for variety of reasons and purposes. More information about IE at UCF can be found at <https://oeas.ucf.edu/index.html>

when M is used in the matrix for institutional effectiveness and other evaluation purposes.

The curriculum map is created to match the universal competencies identified by NASPAA and MPA learning objectives developed by the faculty with significant feedback from advisory board, professional practitioners, students, and alumni. Since the MPA program integrated NASPAA expectations to UCF program review processes and the CM, faculty teaching at the MPA program actively participated and provided feedback. The CM also addressed the existing MPA program strategic plan and objectives with the core courses and helped faculty to examine competencies and make sure none are left unaddressed. The map helped to identify to make sure one competency is addressed by at least one core course in the program. The map helped identify gaps in the program, if any, and potential revisions in each course content. The mapping exercise helps faculty and staff reconsider some of the direct measures used. For example, UCF added a portfolio requirement to the MPA capstone class and included it in the institutional effectiveness measures. A detailed portfolio guide is prepared for the students with courses having embedded direct evaluation strategies to guide faculty engagement and revise course objectives, major project description, and assessment strategies (see Allen, 2004; Donald, 1997; for additional examples).

Faculty members at UCF's MPA program identified intended student learning objectives and outcomes based on NASPAA's required core competencies. A map in the form of a table was created to match the core competencies, learning objectives, and the core courses. The core courses and experiences currently addressing those outcomes and objectives were matched by entering an "I" to indicate students are introduced to the learning objectives; "R" to identify the outcome is reinforced and students are afforded opportunities to practice the competency; and "M" details students have had sufficient practice and can now demonstrate mastery. For simplicity, an "A" can be added to the table indicating where evidence might be collected and evaluated for program-level assessment (collection might occur at the beginning and end of program if comparisons across years are desired).

During academic year 2013–2014, the MPA director facilitated five small group meetings and six school meeting sessions including all MPA faculty and those interested in CM to analyze the competencies and the program learning objectives and make additional revisions if needed. The participants discussed and revised so each outcome is introduced, reinforced, and

mastered. In addition, each outcome should have an "A" to indicate that evidence can be collected for program-level assessment as part of UCF's institutional effectiveness plan. Moreover, a unified syllabus template was created for the MPA program including the core competencies.

The final step in the process was the development of an assessment plan for continuous improvement. This was put in place during 2013–2014 academic year as part of the ongoing UCF IE plan required for each degree program. Ideally, the ongoing assessment is conducted every semester with faculty members reporting on the competencies to be measured in the classes they teach. Faculty report on the grades students achieve for the primary competency taught in each class, as well as, the surveys for graduating students and annual alumni to elicit feedback on the effective utilization of curriculum in teaching and learning (Newcomer, Allen, & Baradei, 2010). The MPA director analyzed the data for each student learning outcome and measure in the IE plan and prepares an annual report to provide the MPA faculty members with feedback on the ability to instill the competencies in students. Faculty members reviewed the findings and considered whether to change the process or the competencies measures for the curriculum.

The competency-based curriculum development and assessment strategies provided an opportunity to revise the MPA program and develop assessment strategies from a holistic perspective. Four steps have been taken to link the competency-based outcomes with activities in the MPA program. First, the UCF Graduating Student Survey was revised based on the NASPAA universal competencies. Likert scale survey questions were added, which is administered by the university's assessment office. Second, in addition to UCF's generic exit survey, the MPA program developed its own graduating student survey designed to include competency domains identified by faculty, advisory council and students in addition to NASPAA's universal competencies completed by graduating students who are taking the capstone course every fall and spring semesters. Third, the IE plan was changed during 2013–2014 to measure the new competency-base curriculum. For each of the five NASPAA Universal Competencies, at least one direct and one indirect measure were determined. The direct measure was always a major assignment from a core course with a grading rubric addressing MPA program competency domains. For the direct measure, achievement was measured through at least 80% or more of students earning a B or higher grade in the major assignment. The indirect measure was one or more questions from the UCF Graduating

Student Survey, with achievement set at 80% or more of the graduating students responding with *agree* or *strongly agree* on their perceived ability to perform NASPAA universal competencies.

Finally, the MPA capstone course was re-designed based on NASPAA core competencies and MPA learning objectives. The course addresses leadership and community building based on students' experience from all core MPA courses, electives, and internships or service learning experiences. The course also requires completion of a professional development portfolio. This core requirement is also included in the graduate catalog as a program requirement. In order to help students develop a professional and usable portfolio, they are provided with a detailed session held during the second week of the semester, an online portfolio preparation webinar, and a sample portfolio. In addition, students are provided with a detailed portfolio guide, the competencies and core courses matrix, and a portfolio assessment rubric at the beginning of each semester.

The goal of the annual IE plan is to evaluate program performance and to make changes to improve the curriculum and the measurement process-closing the loop on the assessment process. The IE plan was revised to include the new portfolio requirement and the new MPA program student survey. In addition to new measurements, the IE process identified inconsistencies in the teaching of competencies between faculty members teaching the same course. The MPA director met with individual faculty members to provide feedback from the first assessment measuring the competencies to address issues of consistency in teaching. Another finding was students were unclear about how certain projects in their portfolios taught the identified competencies. To address this, the MPA program includes the competency-based curriculum map and portfolio requirements in the new student orientation package and in the student handbook.

Lessons learned and recommendations

MPA Programs can build in practice and multiple learning trials for students under three legends as identified in the curriculum map: introduce, reinforce, mastered. Students can perform best if they are introduced to the learning outcome early in the curriculum and then provided sufficient practice and reinforcement before evaluation of their level of mastery takes place as highlighted by Allen (2004).

MPA programs can use the curriculum map to identify the learning opportunities (e.g. assignments, activities) producing the program's outcomes. With the identified

student learning outcomes, the CM can be used as part of institutional effectiveness and outcome assessment strategies. Additionally, the CM allows faculty members to teach to their strengths (each person need not cover all outcomes in a single course). The exercise among the faculty reinforced collaboration and joint responsibility in improving the quality of the MPA program.

MPA programs are able to identify necessary revisions in the program and eliminate outcomes deemed not highly valued and then focus on desired outcomes by including them in multiple courses. The eliminated outcomes can still be course-level outcomes and they need not disappear completely from the curriculum. All the CM participants (students, faculty, staff, alumni, and advisory board) engaged in setting the priorities for the program. This collective action toward common outcomes can be indicators of the successful implementation of the curriculum revisions and the likelihood students will meet or exceed expectations.

The final version of the curriculum map was distributed to students, faculty, and advisory board members. To reinforce the MPA curriculum as a package, each faculty member can make explicit connections across courses for the students. For example, at the beginning of the course, a faculty member can remind students what they were introduced to in another course and explain how the current course will have them practice or expand their knowledge. In the MPA program, the capstone seminar requires completion of a competency-based cumulative portfolio. If the students are introduced to different competencies in each class they can build the portfolio throughout the program.

The CM exercise helped the program director and faculty and to some extent to all others involved in the process a sense of reflective inquiry (shared reflection), shared purpose, shared leadership, in the MPA program. This exercise also helped achieve a coherence in the program. For programs needing to meet NASPAA accreditation requirements, one way to use the CM is to integrate accreditation requirements into the map and examine mission, curriculum, and mandated accreditation standards. The map could serve as a diagnostic tool to identify potential problem areas before accreditation and also be used as a strategic planning tool. This will involve starting from the intended program benchmarks and working back through the courses at different levels to build a path to the objective and mission. The role of external stakeholders, especially the advisory board, should be considered in determining if the students achieved these competencies. The use of resources available at university and college campuses is critical. UCF has an excellent Faculty Center for Teaching and Learning (FCTL) with experts

on measurement and program development. The MPA program faculty had a chance to collaborate with FCTL staff during assessment strategies, CM, and program improvement processes as well.

Conclusion

CM is a beneficial tool, utilized in institutions across disciplines from public health to public administration and across national and international boundaries from Canada to the United Kingdom to Australia, to visually ascertain adherence to competency-based outcomes and assessment. Due to the subjective nature of the process, it is critical for clear definitions to be given in regards to competencies and the desired outcomes. Furthermore, it is imperative to acknowledge the time it takes to complete the mapping process, implement results, and continually assess the content to develop for educational programs.

CM documents what is taught and when, reveals gaps in the curriculum, and helps design an assessment plan. The map helps to improve communication among faculty and program coherence, it also increases the likelihood that students achieve program-level outcomes, and encourages reflective practice. CM provides a method to not only revise and articulate the curriculum, but also a way to foster collaboration and collegiality among those participating in the process. CM needs to be considered as an ongoing, dynamic process and a vehicle for other MPA programs that wish to improve, not only course revisions, but want to promote a collaborative culture enhancing the learning of key stakeholders. It seems that there are not many examples for competency-based CM for MPA programs. Insights into the CM included in this article might be helpful for other MPA programs considering CM based on NASPAA competency-based outcome measures. CM exercised by other disciplines provided some helpful resources for the article. If data collection is adapted based on the curriculum map carefully, this might help streamline evaluation of MPA programs for multiple purposes such as accreditation and institutional state reviews. This article can be beneficial to programs in other disciplines as well.

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References

- Allen, M. J. (2004). *Assessing academic programs in higher education*. Bolton, MA: Anker Publishing.
- American Educational Research Association, American Psychological Association, & National Council on Measurement in Education. (1999). *Standards for educational and psychological testing*. Washington, DC: American Educational Research Association.
- Ashworth, P. D., & Saxton, J. (1990). On competence. *Journal of Further and Higher Education*, 14(2), 3–5. doi:10.1080/0309877900140201
- Awortwi, N. (2010). Building new competencies for government administrators and managers in an era of public sector reforms: The case of Mozambique. *International Review of Administrative Sciences*, 76(4), 723–748. doi:10.1177/0020852310381803
- Baker, H., Brawley, D., Campbell, J., Capozzoli, E., Malgeri, L., & Roberts, G. (2007). Developing a customized program assessment methodology for assurance of learning: A case study. *Journal of College Teaching and Learning*, 4(8), 47–64.
- Barkley, E. F., Cross, K. P., & Major, C. H. (2005). *Collaborative learning techniques: A handbook for college faculty*. San Francisco, CA: Jossey-Bass.
- Bath, D., Smith, C., Stein, S., & Swann, R. (2004). Beyond mapping and embedding graduate attributes: Bringing together quality assurance and action learning to create a validated and living curriculum. *Higher Education Research & Development*, 23(3), 313–328. doi:10.1080/0729436042000235427
- Bawane, J., & Spector, J. (2009). Prioritization of online instructor roles: Implications for competency-based teacher education programs. *Distance Education*, 30(3), 383–397. doi:10.1080/01587910903236536
- Bester, M., & Scholtz, D. (2012). Mapping our way to coherence, alignment and responsiveness. *SAJHE*, 26(2), 282–299.
- Cuevas, N. M., Matveev, A. G., & Miller, K. O. (2010). Mapping general education outcomes in the major: Intentionality and transparency. *Peer Review*, 12(1), 10–15.
- Damrosch, D. (1995). *We scholars: Changing the culture of the university*. Cambridge, MA: Harvard University Press.
- Donald, J. (1997). *Improving the environment for learning*. San Francisco, CA: Jossey-Bass.
- Drake, S. M., & Burns, R. C. (2004). *Meeting standards through integrated curriculum*. Alexandria, VA: Association for Supervision and Curriculum Development.
- Earley, M. A. (2007). Developing a syllabus for a mixed-methods research course. *International Journal of Social Research Methodology*, 10(2), 145–162. doi:10.1080/13645570701334118
- English, F. (1978). *Quality control in curriculum development*. Arlington, VA: American Association of School Administrators.
- Ervin, L., Carter, B., & Robinson, P. (2013). Curriculum mapping: Not as straightforward as it sounds. *Journal of Vocational Education & Training*, 65(3), 309–318. doi:10.1080/13636820.2013.819559
- Glass, K. T. (2007). *Curriculum mapping: A step-by-step guide for creating curriculum year overviews*. Thousand Oaks, CA: Corwin Press.
- Hale, J. A., & Dunlap, R. F., Jr. (Eds.). (2010). *An educational leader's guide to curriculum mapping: Creating and sustaining collaborative cultures*. Thousand Oaks, CA: Sage.

- Hall, G. E., & Hord, S. M. (2006). *Implementing change: Patterns, principles, and potholes*. Boston, MA: Allyn and Bacon.
- Harden, R. M. (2001). Curriculum mapping: A tool for transparent and authentic teaching and learning. *Medical Teacher*, 23(2), 123–137. doi:[10.1080/01421590120036547](https://doi.org/10.1080/01421590120036547)
- Henry, N., Goodsell, C., Lynn, L., Stivers, S., & Wamsley, G. (2009). Understanding excellence in public administration: The report of the task force on educating for excellence in the Master of Public Administration degree for the American Society for Public Administration. *Journal of Public Affairs Education*, 15(2), 117–133.
- Holycross, J. (2006). Curriculum mapping---an essential tool for curriculum development. *The Journal of Physician Assistant Education*, 17(4), 61–64.
- Iglar, K., Whitehead, C., & Takahashi, S. G. (2013). Competency-based education in family medicine. *Medical Teacher*, 35(2), 115–119.
- Jacobs, H. H. (Ed.). (2004). *Getting results with curriculum mapping*. Alexandria, VA: Association for Supervision and Curriculum Development.
- Kapucu, N. (2011). Developing competency-based emergency management degree programs in public affairs and administration. *Journal of Public Affairs Education*, 17(4), 501–521.
- Kapucu, N. (2012). Classrooms as communities of practice: Designing and facilitating learning in a networked environment. *Journal of Public Affairs Education*, 18(3), 585–610.
- Kinchin, I. M. (2011). Visualising knowledge structures in biology: Discipline, curriculum and student understanding. *Journal of Biological Education*, 45(4), 183–189. doi:[10.1080/00219266.2011.598178](https://doi.org/10.1080/00219266.2011.598178)
- Klein, C. J. (2006). Linking competency-based assessment to successful clinical practice. *Journal of Nursing Education*, 45, 379–383.
- Koliba, C. (2004). Assessing reflection assignments for public affairs courses: Implications for educating reflective practitioners. *Journal of Public Affairs Education*, 10(4), 295–309.
- Kuh, G. D., & Ikenberry, S. (2009). *More than you think, less than we need: Learning outcomes assessment in American higher education*. Champaign, IL: National Institute for Learning Outcomes Assessment.
- Litzelman, D. K., & Cottingham, A. H. (2007). The new formal competency based curriculum and informal curriculum at Indiana University School of Medicine: Overview and five- year analysis. *Academic Medicine*, 82, 410–421. doi:[10.1097/ACM.0b013e31803327f3](https://doi.org/10.1097/ACM.0b013e31803327f3)
- Liu, M., Wrobbel, D., & Blankson, I. (2010). Rethinking program assessment through the use of program alignment mapping technique. *Communication Teacher*, 24(4), 238–246. doi:[10.1080/17404622.2010.513002](https://doi.org/10.1080/17404622.2010.513002)
- Martell, K., & Calderon, T. (Eds.). (2005). *Assessment of student learning in business schools: Best practices each step of the way*. Tallahassee, FL: Association for Institutional Research.
- Mau, T. A. (2015). Leadership competencies for a global public service. *International Review of Administrative Sciences*. doi:[10.1177/0020852315576706](https://doi.org/10.1177/0020852315576706)
- Newcomer, K., Allen, H., & Baradei, L. (2010). Improving public service education programs through assessing the performance of MPA alumni. *International Journal of Public Administration*, 33, 311–324. doi:[10.1080/01900690903449681](https://doi.org/10.1080/01900690903449681)
- Raffel, J. A., Maser, S. M., & Calarusse, C. (2012). Accreditation and competencies in education for leadership in public service. In D. C. Menzel, & H. L. White (Eds.), *The state of public administration: Issues, challenges, and opportunities* (pp. 70–88). Armonk, NY: ME Sharpe.
- Robley, W., Whittle, S., & Murdoch-Eaton, D. (2005a). Mapping generic skills curricula: A recommended methodology. *Journal of Further and Higher Education*, 29(3), 221–231. doi:[10.1080/03098770500166801](https://doi.org/10.1080/03098770500166801)
- Robley, W., Whittle, S., & Murdoch-Eaton, D. (2005b). Mapping generic skills curricula: Outcomes and discussion. *Journal of Further and Higher Education*, 29(4), 321–330. doi:[10.1080/03098770500353342](https://doi.org/10.1080/03098770500353342)
- Rubai, N., & Calarusse, C. (2012). Cultural competency as a standard for accreditation. In K. A. Norman-Major, & S. T. Gooden (Eds.), *Cultural competency for public administration* (pp. 219–244). Armonk, NY: ME Sharpe.
- Sandoval, P. A., & Wigle, S. E. (2006). Building a unit assessment system: Creating quality evaluation of candidate performance. *Education*, 126, 640–652.
- Southern Association of Colleges and Schools. (2005). *Resource manual for the Principles of accreditation: Foundations for quality enhancement*. Decatur, GA: SACS-COC.
- Spencer, D., Riddle, M., & Knewstubb, B. (2012). Curriculum mapping to embed graduate capabilities. *Higher Education Research & Development*, 31(2), 217–231. doi:[10.1080/07294360.2011.554387](https://doi.org/10.1080/07294360.2011.554387)
- Suddreth, A., & Galloway, A. T. (2006). Options for planning a course and developing a syllabus. In W. Buskist, & S. F. Davis (Eds.), *Handbook of the teaching of psychology* (pp. 31–35). Malden, MA: Blackwell.
- Sumsion, J., & Goodfellow, J. (2004). Identifying generic skills through curriculum mapping: A critical evaluation. *Higher Education Research & Development*, 23(3), 329–346. doi:[10.1080/0729436042000235436](https://doi.org/10.1080/0729436042000235436)
- Tilbury, D., & Wortman, D. (2008). Education for sustainability in further and higher education: Reflections along the journey. *Planning for Higher Education*, 36(4), 5–16.
- Uchiyama, K. P., & Radin, J. L. (2009). Curriculum mapping in higher education: A vehicle for collaboration. *Innovative Higher Education*, 33(4), 271–280. doi:[10.1007/s10755-008-9078-8](https://doi.org/10.1007/s10755-008-9078-8)
- Udelhofen, S. (Ed.). (2005). *Keys to curriculum mapping: Strategies and tools to make it work*. Thousand Oaks, CA: Corwin Press.
- Wang, L., & Ashcraft, R. F. (2012). Needs assessment and curriculum mapping enhancing management skills of the nonprofit workforce. *Nonprofit Management & Leadership*, 23(1), 121–136. doi:[10.1002/nml.21058](https://doi.org/10.1002/nml.21058)
- Wesp, R., & Miele, J. (2008). Student opinions of the quality of teaching activities poorly predict pedagogical effectiveness. *Teaching of Psychology*, 35(4), 360–362. doi:[10.1080/00986280802374617](https://doi.org/10.1080/00986280802374617)
- Williams, J., & Bolland, K. (2011). Mapping our way to success. *The Journal of Baccalaureate Social Work*, 16(2), 93–114.